CARLA Behavioral-Cloning Teammate Handoff Guide

Manual data collection, project setup, and training handoff

Your role
Drive the CARLA vehicle manually to create clean image-and-steering datasets. The training pipeline is already prepared in the repository; your output is the dataset that it will learn from.

Project goal
Train a behavioral-cloning model that maps a front camera image to a steering command, then evaluate its driving and performance metrics in CARLA.

What is already done

- CARLA 0.9.16 project structure is in place, including the simulator, Python API, source code, dashboard, and model folders.
- Python 3.12 and the CARLA 0.9.16 wheel have been verified on the current laptop.
- Manual control has been tested: the vehicle can be spawned and driven using the pygame window.
- A dedicated manual collector is included at **src/collect_manual_data.py**. It attaches only a front camera to the existing vehicle and does not spawn a duplicate car.
- A previous Traffic Manager/autopilot test was archived under **data/archive_autopilot_test_run_001**. It contains collision/stopped frames and is not training data.

Source of truth
Use the GitHub repository as the shared project. Store every new manual collection run under **data/manual_collected/** and never overwrite an existing run.

1. Set up the project on your laptop

Use a Windows machine if possible, matching the current project setup. The CARLA server must be running before any client script can connect.

Prerequisites

- Git and a GitHub account with access to the repository.
- 64-bit Python 3.12. Use the official Python installer or a Python 3.12 conda environment.
- CARLA 0.9.16 for Windows. CARLA binaries are large and may exceed GitHub file limits, so download the official matching CARLA package separately if the repository does not include the full simulator.
- At least 20 GB free storage; more is better because camera datasets grow quickly.

Clone and install

```
git clone <YOUR-TEAM-REPOSITORY-URL>
cd \"ML-Based Autonomous Car Driving Simulation and Performance Visualization\"

# Create/activate a Python 3.12 environment, then:
python -m pip install -r requirements.txt
python -m pip install PythonAPI/carla/dist/carla-0.9.16-cp312-cp312-win_amd64.whl
```

If a dependency fails
Confirm that **python --version** reports Python 3.12. The Behavior Agent example also needs **shapely**; install it with **python -m pip install shapely** if needed.

Verify the install

```
python -c "import carla; print(carla.__file__)"
```

If this prints a CARLA package path without an error, the Python API is ready.

2. Manual data-collection workflow

Use three separate terminals
This avoids accidentally stopping the CARLA server when you only intend to stop the recorder. Keep all three open while collecting.

Terminal 1 - CARLA server

From the project root, start the simulator and wait for the CARLA window to fully load.

```
.\CarlaUE4.exe
```

Terminal 2 - manual driving

Start CARLA's manual-control client. Click its pygame window before using the keyboard. Use W/A/S/D or the arrow keys to drive.

```
python .\PythonAPI\examples\manual_control.py
```

Terminal 3 - data recorder

Start the manual collector. It finds the existing hero vehicle, attaches one front camera, and writes a new dataset run.

```
python -m src.collect_manual_data --output .\data\manual_collected\run_001
```

Stopping safely
Press **Ctrl+C only in Terminal 3** to finish a collection run. This removes the collector's camera sensor but leaves the manually driven vehicle untouched. Do not press Ctrl+C in Terminal 1 unless you want to close CARLA.

For later sessions, use a new run name: **run_002**, then **run_003**, and so on.

3. How to drive useful training data

The model learns whatever it sees. Smooth, varied, human-controlled examples are much more valuable than many frames of a parked, crashed, or autopilot-controlled car.

Good collection habits

- Make gentle and medium turns in both directions. Avoid holding one steering angle for very long.
- Drive forward at realistic low-to-medium speeds and include starts, smooth braking, and turns at intersections.
- Collect multiple short runs across different roads, weather, and lighting when comfortable.
- Keep the car in lane and use deliberate steering. If you hit an object, stop that run and start a fresh one.
- Aim for several thousand clean moving frames before training; a first small run is useful only as a smoke test.

Avoid

- Do not train from the archived autopilot test run; it contains a collision and many stationary frames.
- Do not reuse a run folder. The recorder deliberately refuses to overwrite it.
- Do not start the recorder before manual_control.py has created its vehicle. If it cannot find the hero vehicle, it will list actor IDs instead of guessing.

Dataset produced by each run

Item	Purpose
images/00000001.png ...	Front-camera frames from the manually driven vehicle.
telemetry.csv	Training-compatible frame, image, steering, throttle, brake, speed, and pose labels.

4. Training handoff and project map

Once manual runs have been reviewed, hand their paths to the training teammate. Training should use only clean runs. The manual recorder writes the same telemetry.csv layout consumed by the existing training pipeline, so a clean run can be used directly with the training command.

Relevant project folders

Path	What it is
src/collect_manual_data.py	Manual recorder created for this project.
src/collect_data.py	Existing Behavior Agent expert-data collector.
src/train.py	Behavioral-cloning training pipeline.
src/drive.py	Run a trained model in CARLA.
data/manual_collected/	New teammate manual-driving runs.
models/	Saved trained .keras models.
logs/	Training metrics and run logs.
app/dashboard.py	Performance dashboard.

Suggested handoff checklist

- Upload source code and documentation to GitHub. Do not commit huge generated datasets, CARLA binaries, models, or logs unless your team intentionally uses Git LFS/cloud storage.
- Share the exact CARLA 0.9.16 version and Python 3.12 requirement with every teammate.
- Record clean manual runs in numbered folders, then review samples before the training teammate uses them.
- Keep a note of the map, weather, date, driver, and any collision or unusual event for every run.

Ready to collect
Start with a short run to confirm that images and labels are saved, then collect longer clean runs. The goal is quality and coverage, not simply the biggest frame count.